SIGMACAMP · ELECTROCHEMISTRY LAB · SESSION 4

Water Electrolysis

Lecture & Instructor Notes

Audience: instructors and teaching assistants

Purpose: Teaching script, deep-dive notes, and coaching guidance

Session hook: Split water and see a 2:1 ratio

Length: 90-minute lab block (~20 min concepts + ~70 min hands-on)

This packet includes

1. Session overview & plan
2. Lecture talking points
3. Instructor deep-dive notes
4. Preparation reminders

Generated from the Electrochemistry workshop curriculum. Prefer the latest PDF from the course repository if procedures change mid-week.

SESSION OVERVIEW & PLAN

Session 4 — Water Electrolysis and Gas Collection

Hook: Split water, collect gases, and see a 2:1 stoichiometric ratio.

Learning outcomes

- Run water electrolysis in a Hoffman-style **U-tube** with graphite electrodes
- Use **sodium sulfate** electrolyte safely (never NaCl)
- Measure H_2 and O_2 volumes and compare to 2:1 ratio
- Connect electrolysis to hydrogen as an energy storage concept
- **Start** electrolytic rust-removal cells for Session 5 reveal

Session plan (90 min)

Time	Activity	Reference
0–10 min	Hook: U-tube cell — which limb fills with gas twice as fast?	lecture.md
10–25 min	Water splitting and stoichiometry	lecture.md
25–40 min	Fill and wire the Na_2SO_4 U-tube apparatus	experiment.md — Part A
40–60 min	Run electrolysis; record volumes over time	experiment.md — Part B
60–70 min	Discuss ratio; optional instructor H_2 pop test	experiment.md — Part C
70–80 min	Brief energy storage discussion	lecture.md
80–90 min	**Start derusting cells** for Session 5	experiment.md — Part E

Key reactions

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Overall: 2H₂O(l) → 2H₂(g) + O₂(g)

Cathode (−): 2H₂O + 2e⁻ → H₂ + 2OH⁻
Anode (+):   2H₂O → O₂ + 4H⁺ + 4e⁻   (acidic media)
          or 4OH⁻ → O₂ + 2H₂O + 4e⁻   (basic media)
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Status

- [ ] U-tube apparatus + graphite electrodes sourced
- [ ] ~0.1 M Na_2SO_4 mixed and labeled
- [ ] Apparatus pilot-tested (2:1 visible)
- [ ] **No saltwater** — Na_2SO_4 only for electrolysis
- [ ] Pop test protocol reviewed
- [ ] Rusty nails + washing soda ready for Part E

LECTURE TALKING POINTS

Session 4 — Lecture: Water Electrolysis and Stoichiometry

Target duration: ~20 minutes (+ energy discussion at end)

> **Instructors:** half-reactions, why no NaCl, 2:1 ratio errors, derusting handoff — instructor.md.

Opening hook (0–10 min)

Show the Hoffman-style **U-tube** with graphite electrodes (or the photo in experiment.md).

"One limb makes hydrogen. One makes oxygen. Which side will fill twice as fast? Why?"

Core concepts (10–25 min)

1. Electrolysis of water

- Pure water conducts poorly → add **sodium sulfate (Na_2SO_4)** as supporting electrolyte (**NOT NaCl**)
- Power supply forces non-spontaneous reaction: water → hydrogen + oxygen

2. Half-reactions (simplified)

Cathode (−): water reduced → **hydrogen gas**
Anode (+): water oxidized → **oxygen gas**

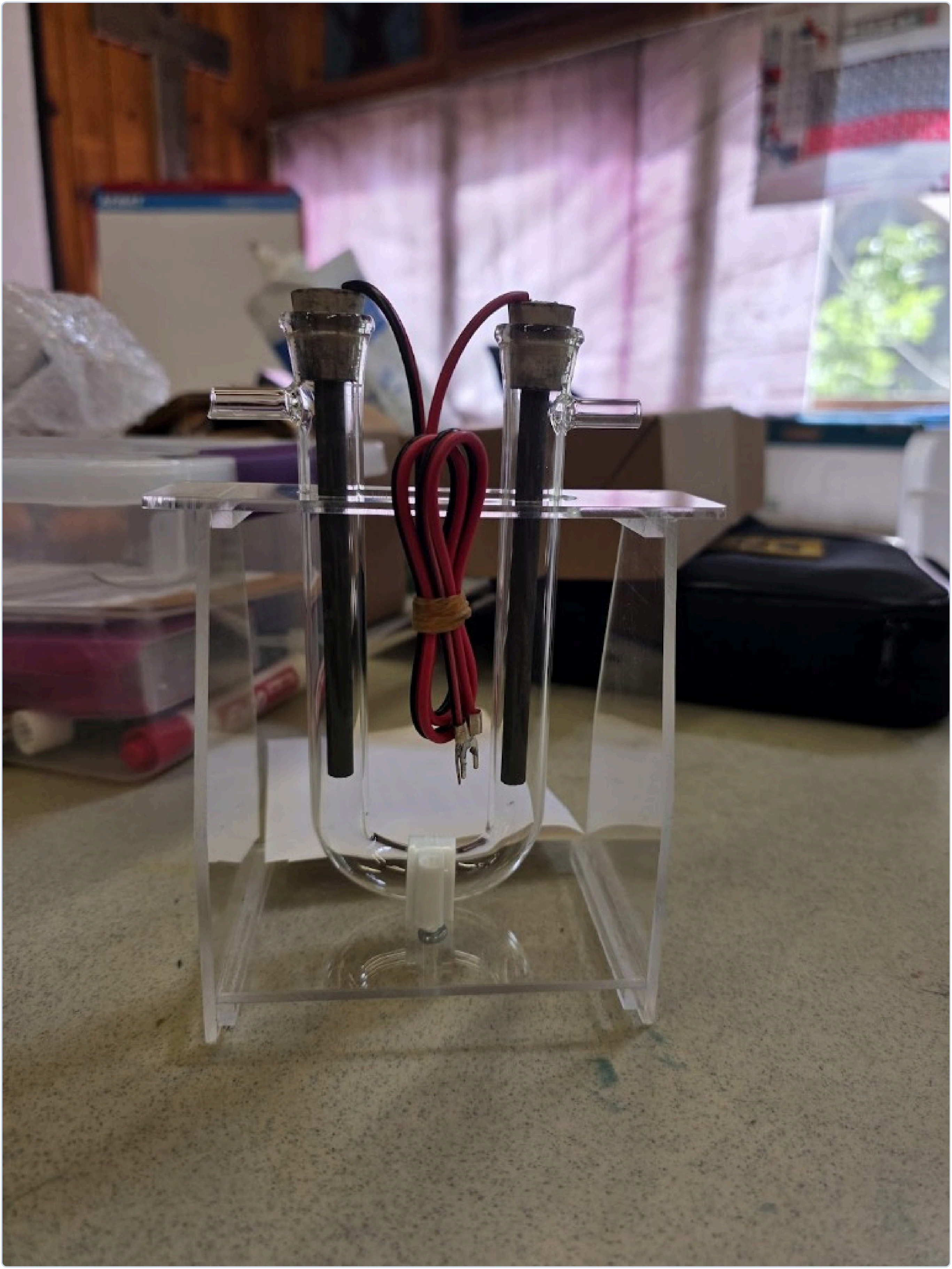
3. Stoichiometry — the 2:1 ratio

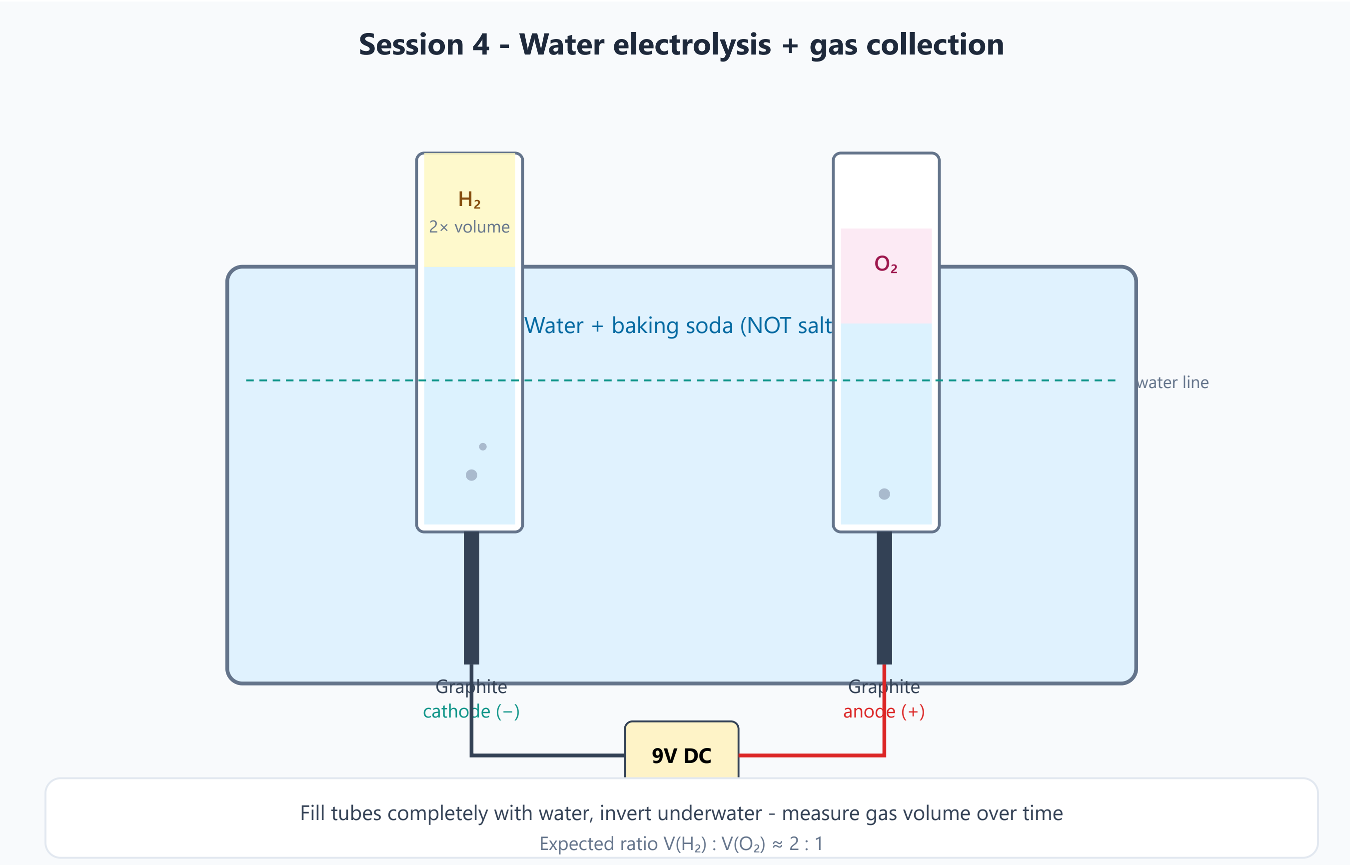
From overall equation: $\mathbf{2H_2O \rightarrow 2H_2 + O_2}$

- **2 volumes H_2 : 1 volume O_2** (same T and P)
- Avogadro: 2 moles H_2 : 1 mole O_2

4. The U-tube cell

- Two graphite electrodes in a glass U-tube (side arms; stoppers at the top)
- Gases collect in each limb; compare column heights / volumes
- Red (+) = anode = O_2; black (−) = cathode = H_2





5. Why observed ratio may differ from 2:1

- O_2 slightly more soluble in water than H_2
- Small bubbles escape through side arms
- Leaks, uneven electrode areas
- Still a strong qualitative and semi-quantitative demo

Energy discussion (75–85 min)

Hydrogen as energy storage

1. **Excess renewable electricity** (solar/wind) → electrolysis → H_2
2. Store H_2 (compressed, or other methods — brief mention)
3. Later: **fuel cell** converts H_2 + O_2 back to electricity + water

Key idea: Hydrogen is an **energy carrier**, not a primary energy source — energy is lost at each step.

Efficiency chain (qualitative)

Solar → electricity → H_2 → storage → fuel cell → electricity → motor/light
(Each arrow loses some energy as heat)

Preview Session 5 (85–90 min)

"We used electricity to make hydrogen. Overnight we'll use electricity to undo rust on iron. Tomorrow: reveal the results, then connect corrosion, galvanic cells, and sacrificial protection."

Start derusting cells per experiment.md Part E before students leave.

Visual aids

- [ ] Balanced equation with mole/volume ratio highlighted
- [ ] Photo of setup with gas volumes marked
- [ ] Simple energy diagram: renewable → H_2 → (optional fuel cell concept)

Vocabulary

- [ ] Electrolysis (applied to water)
- [ ] Stoichiometry / mole ratio
- [ ] Water displacement
- [ ] Energy carrier vs. energy source

Safety talking points (brief in lecture, detailed in experiment)

- **Never use saltwater** — chlorine gas risk
- No open flames near apparatus
- Pop test: instructor only, tiny H_2 volume

INSTRUCTOR DEEP-DIVE NOTES

Session 4 — Instructor Notes: Water Electrolysis Deep Dive

Audience: instructors and TAs only. Pair with lecture.md and experiment.md.

Why this session exists

Students have plated metals. Now electricity splits **molecules** of water into elemental gases, and stoichiometry becomes visible as a **2:1 volume ratio**. This also introduces hydrogen as an energy carrier and sets up overnight electrolytic rust removal for Session 5.

Atomic / molecular foundations

Water's bonding (instructor talking points)

- H_2O: oxygen shares electrons with two hydrogens (polar covalent bonds).
- Oxygen is more electronegative → partial negative charge on O, partial positive on H.
- Liquid water also has extensive hydrogen bonding (affects boiling point, solvation) — optional aside.
- Pure water has very few ions (autoionization): $H_2O \rightleftharpoons H^+ + OH^-$ with tiny $[H^+][OH^-] = 10^{-14}$ at 25 °C. That is why **pure water is a poor conductor**.

Why we add Na_2SO_4

Electrolysis needs mobile ions. **Sodium sulfate (Na_2SO_4)** dissolves into Na^+ and SO_4^{2-} that carry current. It is a **supporting electrolyte** — it enables conduction without (ideally) being the main reactant consumed. This workshop uses **~0.1 M Na_2SO_4** in the U-tube cell.

Baking soda ($NaHCO_3$) is an acceptable emergency backup, but plan on sodium sulfate.

Critical safety contrast:

Electrolyte	OK for classroom gas electrolysis?	Why
Sodium sulfate	**Yes — planned**	Inert supporting electrolyte
Baking soda / $NaHCO_3$	Yes (backup)	No chlorine evolution
Table salt NaCl	**NO**	Chloride oxidizes → **Cl_2 gas** (toxic)

If anyone smells chlorine or sees greenish gas character — **stop immediately**, ventilate, replace electrolyte.

Half-reactions — what to teach vs what is "true"

Classroom-friendly version

Cathode (−):

```
2H₂O + 2e⁻ → H₂(g) + 2OH⁻
```

(or in acid: $2H^+ + 2e^- \rightarrow H_2$)

Anode (+):

$$2H_2O \rightarrow O_2(g) + 4H^+ + 4e^-$$

(or in base: $4OH^- \rightarrow O_2 + 2H_2O + 4e^-$)

Overall:

$$2H_2O(l) \rightarrow 2H_2(g) + O_2(g)$$

Why the 2:1 ratio appears

Oxygen half-reaction involves **4 electrons** per O_2; hydrogen involves **2 electrons** per H_2. Balancing electrons requires twice as many H_2 molecules as O_2 molecules:

$$4e^- \text{ make } 2\ H_2 \text{ and } 1\ O_2 \ \Rightarrow\ V(H_2)/V(O_2) = 2$$

At the same T and P, gas volume ∝ moles (Avogadro / ideal gas). So volumes follow mole ratio.

Electron bookkeeping (Faraday connection)

Same idea as Session 3: charge passed determines moles of e^-, which determines moles of gas — but today students measure **volumes**, not mass.

Thermodynamics (instructor background)

Water splitting is **non-spontaneous** under standard conditions. Rough minimum theoretical voltage for water electrolysis is ~1.23 V (thermodynamic), but real cells need more due to overpotentials and resistance — classroom 9 V packs are plenty. Students need only: *"We must push with a battery; water does not split by itself."*

U-tube apparatus coaching

What students are looking at

Glass U-tube on an acrylic stand; graphite rods through rubber stoppers; side arms near the top of each limb; red/black leads with spade connectors to a DC supply. Fill with **Na_2SO_4**, not salt. Keep side arms open or vented so pressure does not build against sealed stoppers.

Measuring the ratio

Gas collected in each limb displaces liquid downward. Compare gas column heights (or graduated volumes) once bubbling is steady. Equal arm geometry matters for a fair ratio.

Identifying H_2 vs O_2 without a pop test

Clue	H_2 side	O_2 side
Rate	Faster (2× volume)	Slower
Electrode	Cathode (−) / black	Anode (+) / red
Optional	Pop test (instructor)	Relights glowing splint (advanced / careful)

Default classroom ID: **faster = H_2 = cathode**.

Why ratios drift from 2.0

Cause	Effect
O_2 more soluble than H_2	O_2 volume low → ratio high
Leaks / escaped bubbles	Either side low
Unequal electrode area / current distribution	Skewed rates
Air left in tube at $t = 0$	Systematic error
Electrode oxidation / side reactions	Extra gases or lost current

Accept **1.6–2.2** as a successful classroom result; discuss systematics rather than demanding perfection.

Pop test protocol (instructor only)

1. Collect a **tiny** volume of H_2 in a separate small tube.
2. Move away from the main tub and from faces.
3. Brief flame → soft "pop."
4. Never ignite sealed containers or mixed H_2/O_2.
5. Skip entirely if venue rules forbid open flame.

Hydrogen as energy storage (end-of-session talk)

Frame carefully:

1. Renewables make electricity intermittently.
2. Extra electricity can electrolyze water → H_2.
3. H_2 can be stored and later oxidized in a **fuel cell** → electricity + water.
4. H_2 is an **energy carrier**, not a source — the energy came from the electricity (ultimately the sun/wind).
5. Round-trip efficiency < 100% always (heat losses).

You are **not** running a fuel cell demo this week (Session 5 uses rust removal instead). Keep fuel cells as a conceptual bridge only.

Part E — Start electrolytic rust removal (handoff to Session 5)

Correct polarity (students reverse this)

Electrode	Connection	Role
Rusty iron (object to clean)	Battery **−** (cathode)	Reduction / cleaning
Sacrificial steel or graphite	Battery **+** (anode)	Oxidizes / sacrifices
Electrolyte	Washing soda or baking soda	**Not NaCl**

What is happening chemically (simplified)

Rust is largely $Fe_2O_3 \cdot nH_2O$ / FeOOH mixtures. At the cathode, electrochemical reduction and associated chemistry loosen / reduce oxide layers toward metallic iron; H_2 evolution often accompanies the process. The anode corrodes or evolves O_2 depending on material.

Teaching link: same "electricity drives non-spontaneous change" idea as water electrolysis — applied to undoing corrosion.

Overnight logistics

- Label cells; photograph "before."
- Keep a no-power control nail in water.
- Confirm venue policy for leaving batteries connected overnight (prep room preferred).

Misconceptions

Misconception	Correction
"Bubbles are steam."	They are **H_2 and O_2 gases**, not water vapor from boiling.
"Salt makes electrolysis better."	Salt makes **chlorine** — forbidden here.
"Ratio must be exactly 2.000."	Classroom data have solubility and collection errors.
"Hydrogen is a fuel like oil."	It's a **carrier**; energy was put in by electrolysis.

Board plan

1. Write overall $2H_2O \rightarrow 2H_2 + O_2$; circle 2:1.
2. Assign cathode/anode gases with signs.
3. Sketch displacement tubes; emphasize "no air at start."
4. Safety slide: no NaCl.
5. Preview energy carrier concept + start derusting cells before dismissal.

Optional enrichment

- Pourbaix diagrams for Fe–H_2O (why rust forms / removes — advanced).
- Overpotential differences for H_2 vs O_2 on graphite.
- Industrial alkaline electrolyzers vs PEM electrolyzers (names only).

PREPARATION REMINDERS

Session 4 — Preparation

Before class (instructor)

- [ ] **Pilot the U-tube apparatus** — confirm 2:1 ratio visible within session time
- [ ] Check graphite electrodes, stoppers, and spade connectors
- [ ] Mix **~0.1 M Na_2SO_4** batch; label **"NO SALT"**
- [ ] Review pop test safety; decide if demo will run (weather, room rules)
- [ ] Optional: test solar panel output with the same cell
- [ ] Stage rusty nails, washing soda, sacrificial anodes, and spare jars for **Part E** (end of class)
- [ ] Confirm overnight power policy for derusting cells (prep room vs. disconnect)

Day-of setup

- [ ] One U-tube station per group (or shared stations if kits are limited)
- [ ] Pre-mix Na_2SO_4; aliquot or fill stations
- [ ] Large sign: **"No salt in electrolysis — use sodium sulfate"**
- [ ] Separate instructor area for pop test marked off
- [ ] Part E station: rusty nails, washing soda, steel/graphite anodes, labels, camera

Common failure modes (from pilot)

Issue	Prevention
Weak bubble rate	Fresh ~0.1 M Na_2SO_4; clean graphite; firm spade connections
Students add salt "to speed up"	Pre-brief; labeled Na_2SO_4 only
Overpressure	Keep side arms open or vented
Reversed polarity	Mark (−)=H_2 / (+)=O_2 on the stand before power-on

Open questions / notes

- Supply voltage used:
- Best volume-measurement method (graduations vs. ruler):
- Pop test yes/no for this venue: